

The solution you opt for is also decided by your budget and the level of protection you require for your hardware.

When combatting power problems, there are several weapons with which you can arm your sensitive electronics components.

■ **Circuit breaker:** This device is a switch capable of responding to dangerous overload levels in an electrical circuit. When installed, the switch detects if the current level of a line is too high and instantaneously (in tens of milliseconds) 'trips' and switches off. This effectively disconnects the circuit it's connected to and prevents the equipment from receiving dangerously high input power levels. However, since it effectively switches off the power, it's a rather quick-and-dirty way to implement power protection.

Circuit breakers are particularly useful if you want to protect your computer from damage due to failure or malfunction of other equipment that is on the same circuit as your computer. The circuit breaker does not provide any backup power, nor does it regulate the voltage reaching your equipment. For good power protection, this system should be implemented in conjunction with one or more of the methods given below.

■ **Spike buster:** This device comes almost as standard with new computer purchases. A spike buster (also called a surge suppresser) looks similar to a power strip that provides added power outlets, except that it contains circuitry that can detect impulse increases (or spikes) in the input voltage. When it detects a higher voltage, it shuts off and its fuse opens. This prevents a dangerously high voltage from reaching your equipment. Such devices can protect your computer in case of errors in power lines or even lightning strikes, which can cause sharp voltage increases. However, in case of a power failure, a spike buster will not be able to power your equipment, hence it is not the best way to implement power protection.

■ **Stabiliser:** A stabiliser is a slightly better device for power protection—it's capable of detecting both power surges and brownouts and can rectify them to deliver a regulated AC voltage to your computer. It continuously senses the input voltage (and in some cases even the output voltage). It uses a closed

AD 12m x 25 cm